

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – C Code Output Prediction

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 1 – C Code Output Prediction
Weightage:	35% of CIA	Total Marks:	100
CO1 Statement:	<i>Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.</i>		
Student Name:	R.Paul David	Reg. No.:	192572144
Section / Batch:	CSE(AI)	Date:	08/07/2026

Code of Conduct

I R.Paul David(192572144) (Name / Reg No)
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: R.Paul David

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Linked data structure	Basic data structure	1 - 2	20
Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure	Classifications of Linear data structure	3 - 4	20
Search Data Structure	Linear Search and Binary Search	5 - 6	20
Recursion, Performance analysis	Recursion	7 - 8	20
Array Data Structures	Implementations Using Array data structure	9 - 10	20
GRAND TOTAL:			100 Marks

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QUESTIONS

Q1

C Code - Pointer with Array

10 marks

✓

Q2

C Code - Linked List Node

10 marks

✓

Q3

C Code - Binary Search Complexity

10 marks

✓

Q4

C Code - Pointer Increment

10 marks

✓

Q5

C Code - Array Address

10 marks

✓

Q6

C Code - Linked List Traversal

10 marks

✓

Q1 C Code - Pointer with Array

10 marks

```
#include<stdio.h>
int main(){
int arr[]={10,20,30};
int *p=arr;
printf("%d",*(p+2));
}
```

Your Answer

30

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QUESTIONS

Q1

C Code - Pointer with Array

10 marks

✓

Q2

C Code - Linked List Node

10 marks

✓

Q3

C Code - Binary Search Complexity

10 marks

✓

Q4

C Code - Pointer Increment

10 marks

✓

Q5

C Code - Array Address

10 marks

✓

Q6

C Code - Linked List Traversal

10 marks

✓

Q3 C Code - Binary Search Complexity

10 marks

```
while(low<=high){
mid=(low+high)/2;
}
```

Your Answer

0(log n)

Save Answer

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10 marks

Q6 

C Code – Linked List Traversal

10 marks

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Q6
C.Code – Linked List Traversal
10 marks

10 marks

Your Answer

10

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QUESTIONS

Q1

C Code - Pointer with Array

10 marks

Q2

C Code - Linked List Node

10 marks

Q3

C Code - Binary Search Complexity

10 marks

Q4

C Code - Pointer Increment

10 marks

Q5

C Code - Array Address

10 marks

Q6

C Code - Linked List Traversal

10 marks

Q7

C Code - Recursion (Factorial)

10 marks

Q7 C Code - Recursion (Factorial)

10 marks

```
#include <stdio.h>

int fact(int n){
    if(n==0)
        return 1;
    return n * fact(n-1);
}

int main(){
    printf("%d", fact(4));
    return 0;
}
```

Your Answer

24

Save Answer

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10 marks

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Q6
C Code – Linked List Traversal
10 marks

10 marks

 Save Answer

CO1_AT1_C CODE OUTPUT PREDICTION

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QUESTIONS

Q1 ✓
C Code - Pointer with Array
10 marks

Q2 ✓
C Code - Linked List Node
10 marks

Q3 ✓
C Code - Binary Search
Complexity
10 marks

Q4 ✓
C Code - Pointer Increment
10 marks

Q5 ✓
C Code - Array Address
10 marks

Q6 ✓
C Code - Linked List Traversal
10 marks

Q10 C Code – Linear Data Structure (Array)

10 marks

```
#include <stdio.h>
int main()
{
    int arr[]={2,4,6,8};
    printf("%d",arr[3]);
}
```

Your Answer

8

 **Save Answer**